Project Title: Smart Automatic Water Tank Management and IoT Monitoring System

Overview of The Project:

- Imagine a large apartment, industry, college campus, or hospital where hundreds of people depend on a water tank for their daily water supply. Traditionally, a person must manually check the water level every few hours and switch the water pump ON or OFF. Sometimes the pump continues running even after the tank becomes full, causing water overflow and wastage. At other times, the tank becomes empty because nobody notices the low water level, resulting in an interruption of the water supply. To overcome these problems, we designed the **Smart Automatic Water Tank Management and IoT Monitoring System** using the **LPC2129 ARM7 Microcontroller**.
- The entire system is powered by a **5V DC Power Supply**, which provides the required operating voltage to all sensors, communication modules, storage devices, and output devices. The **LPC2129 ARM7** acts as the brain of the system. It continuously collects information from different sensors, processes the received data, makes intelligent decisions, controls the water pump automatically, stores important information, and uploads the system status to the cloud.
- The system continuously monitors the amount of water available inside the overhead tank using the **Water Level**

Sensor. The sensor measures the water level and sends the information to the LPC2129 through the ADC channel. Based on this value, the controller determines whether the tank is empty, partially filled, or completely full.

- At the same time, the **Ultrasonic Sensor** measures the distance between the sensor and the surface of the water. This provides another method of calculating the water level inside the tank. By comparing both measurements, the controller can obtain more reliable information about the actual water level. If any abnormal reading is detected, the system can identify possible sensor faults or unexpected conditions.
- The **LM35 Temperature Sensor** continuously measures the temperature of the surrounding environment or water. Temperature information is useful because very high temperatures may affect the water storage system or indicate abnormal environmental conditions. The measured temperature is displayed on the LCD and can also be uploaded to the cloud for remote monitoring.
- The system also includes **User Switches**, allowing the operator to manually control the water pump whenever required. During maintenance or emergency situations, the operator can switch between automatic mode and manual mode without modifying the software. This provides greater flexibility for system operation.
- Every important event occurring inside the system is recorded using the **DS1307 Real Time Clock (RTC)**. The RTC continuously provides the current date and time to the LPC2129. Whenever the pump starts, stops, the tank

becomes full, or the water level becomes critically low, the exact time is recorded. This allows administrators to analyze the daily operation of the water management system.

- Based on the information received from all the sensors, the **LPC2129 ARM7** makes intelligent decisions. If the water level falls below the minimum threshold, the controller automatically turns ON the water pump. As water is pumped into the tank, the controller continuously monitors the changing water level. When the tank reaches the maximum level, the controller immediately turns OFF the water pump, preventing unnecessary water overflow and reducing electricity consumption.
- The **Relay Driver** acts as an electronic switch between the LPC2129 and the water pump. Since the microcontroller cannot directly drive a high-power motor, it activates the relay, which safely controls the pump. The relay ensures reliable switching while protecting the controller from high voltage and current.
- The system also includes a **Servo Motor**, which controls the water valve automatically. Depending on the operating conditions, the LPC2129 rotates the servo motor to open or close the valve, allowing automatic regulation of water flow without human intervention.
- A **16×2 LCD Display** continuously shows the operating status of the entire system. Important information such as current water level, temperature, pump status, valve position, and system messages are displayed in real time,

allowing the operator to monitor the complete system without connecting additional equipment.

- Visual and audible indications are provided using **LED Indicators** and a **Buzzer**. The LEDs indicate different operating conditions such as normal operation, low water level, or tank full status. Whenever an abnormal condition occurs, such as critically low water level or sensor failure, the buzzer alerts the operator immediately.
- To enable remote monitoring, the system uses the **ESP-01 Wi-Fi Module**. The LPC2129 communicates with the ESP-01 through UART and uploads important information such as water level, pump status, valve status, temperature, and alarm conditions to the ThingSpeak cloud platform. This allows the user to monitor the entire water tank system remotely using a computer or mobile phone from anywhere with Internet access.
- The project also stores important information in two different memory devices. The **AT24C256 EEPROM** stores permanent configuration data and system parameters that must be retained even after power failure. The **SD Card Module** stores detailed operational logs such as pump ON/OFF timings, water level history, temperature records, and alarm events. These records can later be analyzed to understand water usage patterns and improve system efficiency.
- For large buildings or industrial environments, the system supports communication with multiple controllers using the **MCP2551 CAN Transceiver**. If several water tanks are installed at different locations, each tank can have its own

LPC2129 controller. Through the CAN network, all controllers exchange information, allowing centralized monitoring and coordinated operation of the complete water distribution system.

- By integrating sensing, automatic control, data logging, IoT connectivity, and industrial communication, the **Smart Automatic Water Tank Management and IoT Monitoring System** provides a complete real-time solution for intelligent water management. The project minimizes water wastage, reduces electricity consumption, eliminates continuous human supervision, maintains accurate operational records, and enables remote monitoring, making it suitable for apartments, industries, hospitals, educational institutions, commercial buildings, and smart city water management applications.

Project Explanation:

Water is one of the most valuable natural resources, and efficient water management is essential in homes, apartments, schools, hospitals, industries, and commercial buildings. In many places, water tanks are still operated manually, which may lead to water overflow, unnecessary pump operation, increased

electricity consumption, and higher maintenance.

The Smart Automatic Water Tank Management and IoT Monitoring System is designed using the LPC2129 ARM7 Microcontroller to automate the complete water tank monitoring process. The system continuously monitors the water level using the Water Level Sensor and Ultrasonic Sensor, while the LM35 Temperature Sensor measures the surrounding temperature. Based on these inputs, the LPC2129 automatically controls the Relay, which switches the Water Pump ON or OFF, and operates the Servo Motor to control the water valve whenever required.

The current system information is displayed on the 16×2 LCD and simultaneously transmitted through UART for monitoring and debugging. Important operational data is stored in the AT24C256 EEPROM, and real-time information is uploaded to the ThingSpeak cloud using the ESP-01 Wi-Fi Module. The DS1307 RTC provides accurate date and time for event recording, while the MCP2551 CAN Transceiver enables communication with other CAN-enabled embedded systems.

By integrating automatic monitoring, intelligent control, local display, data storage, cloud connectivity, and industrial communication into a single embedded platform, this project provides a reliable and modern solution for smart water tank management.

Introduction

Water management plays an important role in reducing water wastage and improving system efficiency. Manual operation of water tanks requires continuous monitoring and may result in overflow, dry running of pumps, and unnecessary power consumption.

The Smart Automatic Water Tank Management and IoT Monitoring System uses the LPC2129 ARM7 Microcontroller to monitor water level, measure temperature, control the water pump, and operate the water valve automatically. The system displays real-time information on the LCD, sends data through UART, stores important records in the EEPROM, and uploads the system status to the ThingSpeak cloud using the ESP-01 Wi-Fi Module. This combination

provides an intelligent, reliable, and easy-to-monitor water management solution.

Objective of the Project

- To automatically monitor the water level inside the storage tank.
- To measure the distance using the Ultrasonic Sensor.
- To monitor surrounding temperature using the LM35 Temperature Sensor.
- To automatically control the water pump using a Relay.
- To control the water valve using a Servo Motor.
- To display the current system status on the LCD.
- To provide real-time system information through UART.
- To store important system information in the EEPROM.
- To upload water tank information to the ThingSpeak cloud using the ESP-01 Wi-Fi Module.
- To maintain accurate date and time using the DS1307 RTC.
- To support CAN communication for future industrial applications.

- To reduce water wastage, electricity consumption, and manual effort.

Need of the Project

Modern water management systems require automatic monitoring, intelligent control, and remote accessibility. Continuous manual observation is time-consuming and may result in water overflow, unnecessary pump operation, and increased maintenance costs.

This project is designed to provide automatic monitoring of the water tank, intelligent pump control, and real-time system information. The system displays the current status on the LCD, provides monitoring through UART, stores important operational data in the EEPROM, and uploads real-time information to the ThingSpeak cloud. These features improve reliability, simplify monitoring, and reduce human intervention.

Proposed System

The proposed system is built around the LPC2129 ARM7 Microcontroller, which acts as the central controller for the complete water tank management

system.

The controller continuously receives information from the Water Level Sensor, Ultrasonic Sensor, LM35 Temperature Sensor, User Switches, and DS1307 RTC. Based on the received information, it automatically controls the Relay, Water Pump, Servo Motor, LCD, LEDs, and Buzzer.

The current system status is displayed on the LCD, transmitted through UART, stored in the AT24C256 EEPROM, and uploaded to the ThingSpeak cloud through the ESP-01 Wi-Fi Module. The MCP2551 CAN Transceiver enables communication with other CAN-enabled devices for future industrial automation and distributed monitoring.

Features of Proposed System

- Automatic water level monitoring.
- Automatic water pump control.
- Ultrasonic-based water level measurement.
- Temperature monitoring using the LM35 sensor.
- Servo-controlled water valve operation.
- Real-time LCD status display.

- UART-based system monitoring and debugging.
- EEPROM-based storage of important system information.
- ThingSpeak cloud monitoring through the ESP-01 Wi-Fi Module.
- RTC-based date and time management.
- LED and Buzzer indication for warning conditions.
- CAN communication support for future expansion.
- Low power consumption.
- Reduced manual operation.
- Reliable and intelligent embedded system.

Main Concept of the Project

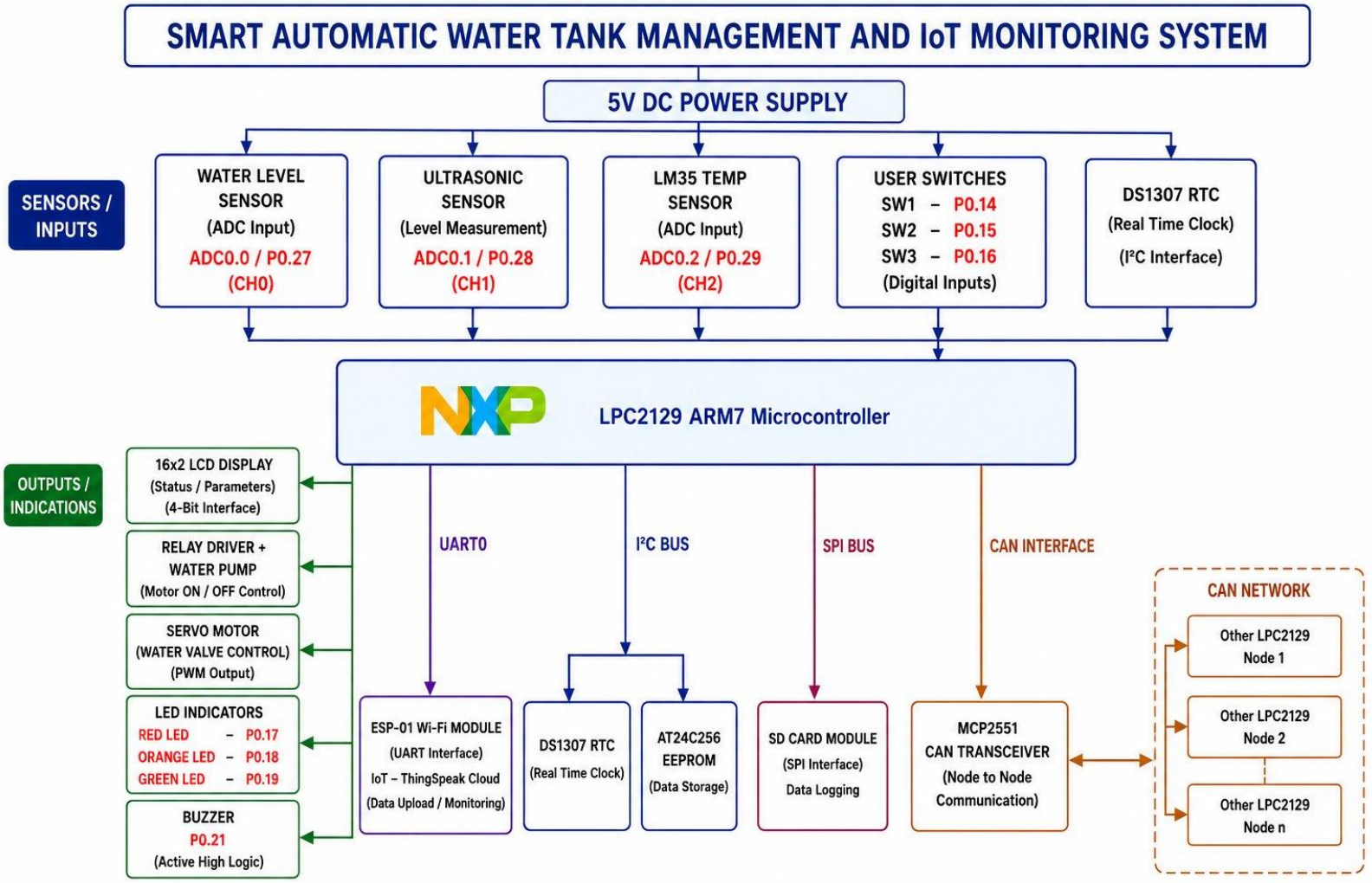
The main concept of this project is to develop an automatic and intelligent water tank management system using the LPC2129 ARM7 Microcontroller.

The controller continuously collects information from the Water Level Sensor, Ultrasonic Sensor, LM35 Temperature Sensor, User Switches, and DS1307 RTC. Based on the received information, it automatically controls the Relay, Water Pump, and Servo Motor to maintain the required water level.

The processed information is displayed on the LCD, transmitted through UART, stored in the EEPROM, and uploaded to the ThingSpeak cloud through the ESP-01 Wi-Fi Module. The MCP2551 CAN Transceiver provides future support for CAN-based industrial communication.

This combination of sensing, automatic control, local display, data storage, cloud connectivity, and industrial communication provides a complete smart water tank management solution.

Block Diagram :



Working Principle

The Smart Automatic Water Tank Management and IoT Monitoring System continuously monitors the water level and surrounding conditions using multiple sensors. The LPC2129 ARM7 Microcontroller receives information from the Water Level Sensor, Ultrasonic Sensor, LM35 Temperature Sensor, User Switches, and DS1307 RTC. After processing the received data, it automatically controls the Relay, Water Pump, Servo Motor, LCD Display, LEDs, and Buzzer. The measured information is displayed on the LCD, transmitted through UART, stored in the EEPROM, and uploaded to the ThingSpeak cloud using the ESP-01 Wi-Fi Module. The system continuously repeats this process to provide reliable and automatic water tank management.

System Architecture

Input Devices

- Water Level Sensor
- Ultrasonic Sensor
- LM35 Temperature Sensor
- User Switches
- DS1307 RTC

Processing Unit

- LPC2129 ARM7 Microcontroller

Communication & Storage

- ESP-01 Wi-Fi Module
- AT24C256 EEPROM
- SD Card Module
- MCP2551 CAN Transceiver

Output Devices

- LCD Display
- Relay Module
- Water Pump
- Servo Motor
- LEDs
- Buzzer

Power Supply

- Regulated DC Power Supply

Hardware Description + Expected Outputs

LPC2129 ARM7 Microcontroller

The LPC2129 ARM7 Microcontroller is the heart of the complete project. It receives data from all input devices, processes the information, and controls every output module. The controller communicates with

peripherals through UART, I²C, SPI, and GPIO interfaces. It continuously updates the LCD, controls the relay and servo motor, and exchanges data with external communication modules. It also coordinates storage, cloud updates, and overall system operation.

UART Output

System : Running

Controller : LPC2129

Status : Active

LCD Output

System Ready

Status : ON

LED Indication

🟢 Green LED : ON

🟡 Yellow LED : OFF

🔴 Red LED : OFF

Water Level Sensor

The Water Level Sensor continuously measures the amount of water available inside the storage tank. The measured value is sent to the LPC2129 for processing.

The controller compares the level with predefined limits to decide whether the pump should be turned ON or OFF. This automatic monitoring prevents water overflow and avoids dry running of the pump. The current water level is displayed on the LCD and transmitted through UART for monitoring.

UART Output

Water Level : 35%

Tank Status : LOW

Pump Status : ON

LCD Output

Water : 35%

Pump : ON

LED Indication

Green LED : OFF

Yellow LED : ON

Red LED : OFF

Ultrasonic Sensor

The Ultrasonic Sensor measures the distance between the sensor and the water surface using ultrasonic waves. The measured distance is converted into the corresponding water level by the LPC2129. This method provides accurate level measurement and improves automatic pump control. The controller continuously updates the measured distance whenever the water level changes. The information is available on both the LCD and UART terminal.

UART Output

Distance : 18 cm

Water : Medium

Status : NORMAL

LCD Output

Dist :18 cm

Level : MED

LED Indication

Green LED : ON

Yellow LED : OFF

Red LED : OFF

LM35 Temperature Sensor

The LM35 Temperature Sensor measures the surrounding temperature near the water tank. It provides an analog voltage proportional to temperature, which is converted into digital form by the LPC2129 ADC. The measured temperature helps monitor environmental conditions around the system. The value is displayed on the LCD and transmitted through UART. This feature can also be extended for industrial monitoring applications.

UART Output

Temperature : 30°C

Status : NORMAL

LCD Output

Temp :30°C

Status : NORMAL

LED Indication

- Green LED : ON
- Yellow LED : OFF
- Red LED : OFF

User Switches

The User Switches allow manual operation of the system whenever required. They are useful during testing, maintenance, and emergency conditions. The user can manually start or stop the pump and verify different hardware modules. The LPC2129 continuously monitors the switch status and executes the corresponding action immediately. This provides flexibility along with automatic operation.

UART Output

Switch : SW1

Action : Pump ON

LCD Output

Manual Mode

Pump : ON

LED Indication

- Green LED : ON
- Yellow LED : OFF
- Red LED : OFF

LCD Display

The 16×2 LCD Display provides real-time information about the complete water tank management system. It displays water level, temperature, pump status, and other important messages. The LCD enables users to monitor the system without connecting a computer. It also displays warning and status information during normal operation. This improves usability and simplifies troubleshooting.

UART Output

LCD Updated

Display Active

LCD Output

Water : 80%

Pump : OFF

LED Indication

Green LED : ON

Yellow LED : OFF

Red LED : OFF

Hardware Description + Expected Outputs

ESP-01 Wi-Fi Module

The ESP-01 Wi-Fi Module provides wireless internet connectivity for the complete water tank management system. It communicates with the LPC2129 through the UART interface and transfers sensor data to the ThingSpeak cloud. This enables users to monitor the system remotely using the internet. The module uploads water level, temperature, and pump status whenever new data is available. It reduces manual monitoring and makes the project suitable for IoT applications.

UART Output

Wi-Fi : Connected

Server : ThingSpeak

Upload : Success

LCD Output

WiFi : Connected

Cloud : Updated

LED Indication

- Green LED : ON
- Yellow LED : OFF
- Red LED : OFF

DS1307 RTC

The DS1307 Real-Time Clock maintains the current date and time even when the main power is removed. It communicates with the LPC2129 using the I²C interface and provides accurate timestamps for system events. The RTC records the timing of pump operations and sensor updates. It improves event logging and supports future scheduling features. The clock continues running using its backup battery.

UART Output

Date : 30-07-2026

Time : 10:45:30

LCD Output

30/07/26

10:45 AM

LED Indication

- Green LED : ON

Yellow LED : OFF

Red LED : OFF

AT24C256 EEPROM

The AT24C256 EEPROM permanently stores important project data even after power is switched OFF. The LPC2129 reads and writes data through the I²C interface. Water level, temperature, pump status, and event records can be saved for future reference. This storage helps in data recovery and troubleshooting. The memory contents remain available until they are intentionally erased.

UART Output

EEPROM : Write OK

Memory : Updated

LCD Output

Data Saved

EEPROM OK

LED Indication

Green LED : ON

Yellow LED : OFF

🔴 Red LED : OFF

SD Card Module

The SD Card Module provides large-capacity storage for long-term data logging. It stores water level history, temperature records, pump activity, and alarm information in files. The stored data can be transferred to a computer for analysis. It is useful where EEPROM capacity is insufficient. The module makes the project suitable for industrial monitoring.

UART Output

SD Card : Ready

Log Saved

LCD Output

SD : Active

Data Saved

LED Indication

🟢 Green LED : ON

🟡 Yellow LED : OFF

🔴 Red LED : OFF

MCP2551 CAN Transceiver

The MCP2551 CAN Transceiver enables reliable communication with other CAN-enabled embedded systems. It converts the controller CAN signals into differential bus signals for long-distance communication. It supports industrial automation and multi-node networking. CAN communication is highly reliable even in noisy environments. The module allows future system expansion.

UART Output

CAN : Active

Frame : Sent

LCD Output

CAN : OK

Network ON

LED Indication

Green LED : ON

Yellow LED : OFF

Red LED : OFF

LEDs

The LEDs provide immediate visual indication of the operating status of the system. Different LEDs indicate normal operation, warning conditions, and fault conditions. They help users quickly understand the system status without checking the LCD. LEDs are useful during debugging and demonstrations. They provide a simple and reliable indication mechanism.

UART Output

LED : GREEN

Status : NORMAL

LCD Output

System OK

Normal Mode

LED Indication

Green LED : ON

Yellow LED : OFF

Red LED : OFF

Buzzer

The Buzzer generates an audible alert whenever abnormal conditions are detected. It warns the user

about low water level, overflow, communication failure, or other fault conditions. The buzzer immediately attracts attention even if the display is not being observed. It improves the safety and reliability of the project. It is also useful during testing and maintenance.

UART Output

Alarm : ON

Buzzer : Active

LCD Output

Warning !!

Check Tank

LED Indication

Green LED : OFF

Yellow LED : OFF

Red LED : ON

11. Communication Interfaces

The project uses multiple communication interfaces to connect sensors, storage devices, and communication modules with the LPC2129 ARM7 Microcontroller.

- UART – ESP-01 Wi-Fi Module and PC serial monitoring.
- I²C – DS1307 RTC and AT24C256 EEPROM.
- SPI – SD Card Module.
- CAN – MCP2551 CAN communication.
- GPIO – Relay, LEDs, Buzzer, Switches and Sensors.

Final Expected Output on Serial Terminal (UART0)

```
==========================================
 SMART WATER TANK MANAGEMENT SYSTEM
==========================================

Controller      : LPC2129 ARM7
Project Status  : RUNNING


Water Level     : 78%
Distance        : 12 cm
Temperature     : 29°C

Pump Status     : OFF
Relay Status    : OFF
```

Valve Status : CLOSED

Wi-Fi Status : CONNECTED
Cloud Status : UPDATED

RTC Date : 30/07/2026
RTC Time : 10:45:30 AM

EEPROM Status : DATA SAVED
SD Card Status : LOG SAVED
CAN Status : ACTIVE

System Status : NORMAL

Monitoring...

If Water Level is Low

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SMART WATER TANK MANAGEMENT SYSTEM

==========

Controller : LPC2129 ARM7

Project Status : RUNNING

Water Level : 20%

Distance : 45 cm

Temperature : 31°C

Pump Status : ON

Relay Status : ON

Valve Status : OPEN

Wi-Fi Status : CONNECTED

Cloud Status : UPDATED

RTC Date : 30/07/2026

RTC Time : 11:15:10 AM

EEPROM Status : DATA SAVED

SD Card Status : LOG SAVED

CAN Status : ACTIVE

Warning : LOW WATER LEVEL

Monitoring...

If Tank is Full

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SMART WATER TANK MANAGEMENT SYSTEM

==========

Controller : LPC2129 ARM7

Project Status : RUNNING

Water Level   : 100%

Distance   : 2 cm

Temperature   : 30°C

Pump Status   : OFF

Relay Status   : OFF

Valve Status   : CLOSED

Wi-Fi Status   : CONNECTED

Cloud Status   : UPDATED

RTC Date   : 30/07/2026

RTC Time   : 12:05:20 PM

EEPROM Status   : DATA SAVED

SD Card Status : LOG SAVED

CAN Status : ACTIVE

Message : TANK FULL

Monitoring...

All The Best…